

Comprehensive Education CRM and Student Application Management Platform

The proposed system should be a **multi-tenant, role-based Education CRM** for managing student leads, counselling, university applications, documents, partner universities, external agents, communication, reporting, and operational performance.

1. System Objectives

The web application should enable education consultants, universities, recruitment agencies, counsellors, students, and external agents to:

- Capture and manage prospective-student leads.
- Convert leads into university applications.
- Track applications through configurable workflows.
- Manage student documents and compliance requirements.
- Maintain university, campus, programme, intake, fee, and eligibility data.
- Coordinate work among counsellors, agents, offices, and teams.
- Communicate with students through email, WhatsApp, SMS, and internal messaging.
- Use AI to analyse documents, identify missing information, recommend programmes, and prepare applications.
- Generate operational, financial, conversion, and performance reports.
- Maintain security, privacy, auditability, and regulatory compliance.

2. User Roles

The system should support configurable roles rather than relying only on fixed roles.

Role	Main Responsibilities
Platform Super Admin	Manage tenants, subscriptions, system settings, security, global configuration, and platform monitoring
Organization Admin	Configure the organization, offices, users, roles, workflows, integrations, and master data
Office Manager	Manage a branch office, counsellors, leads, workloads, and performance
Team Leader	Assign applications and monitor team activity
Counsellor	Manage leads, students, documents, applications, communication, and follow-ups

Admissions Officer	Review applications, verify documents, and update application decisions
Compliance Officer	Verify identity, finances, academic documents, and regulatory compliance
Finance Officer	Manage fees, invoices, commissions, refunds, and payments
External Agent	Submit and track applications for referred students
University Partner	Review submitted applications and update decisions where portal access is enabled
Student/Applicant	Complete profile, upload documents, select programmes, communicate, and track application progress
Auditor/Read-only User	Review records, reports, actions, and audit history without editing
Support User	Handle technical and operational support cases

A user may hold multiple roles across different offices, teams, or organizations.

3. Functional Requirements

3.1 Authentication and Account Management

The system shall:

1. Allow users to register through invitation, self-registration, agent onboarding, or student signup.
2. Support login using:
 - o Email and password
 - o Google
 - o Microsoft
 - o University or organization SSO
3. Support multi-factor authentication.
4. Support password reset and account recovery.
5. Lock accounts after configurable failed login attempts.
6. Allow admins to activate, suspend, deactivate, or archive users.
7. Maintain login history, session history, IP address, browser, and device information.
8. Allow users to view and terminate active sessions.

9. Support configurable password policies.
10. Support session timeout and automatic logout.
11. Require acceptance of privacy policy, terms, and consent notices.
12. Support user impersonation by authorized platform administrators with full audit logging.

3.2 Multi-Tenant Organization Management

The system shall:

1. Support multiple independent organizations on the same platform.
2. Isolate each organization's users, students, applications, documents, reports, and settings.
3. Allow each organization to configure:
 - o Name
 - o Logo
 - o Theme
 - o Domain or subdomain
 - o Contact details
 - o Time zone
 - o Currency
 - o Language
 - o Date format
 - o Academic cycle
4. Support multiple branch offices.
5. Allow one office to be marked as head office.
6. Assign users to one or multiple offices.
7. Allow organization-specific terms, privacy policies, email templates, and workflows.
8. Support custom branding and white-labelling.
9. Allow organizations to define business hours and holidays.
10. Maintain subscription plan, usage limits, storage, users, applications, and AI credits.

3.3 Role and Permission Management

The system shall:

1. Provide predefined roles.
2. Allow custom roles to be created.
3. Support permissions at:
 - o Module level
 - o Screen level
 - o Record level
 - o Field level
 - o Action level
4. Control permissions for create, read, update, delete, export, import, approve, assign, and archive.
5. Restrict users to:
 - o Assigned office
 - o Assigned team
 - o Assigned applications
 - o Assigned countries
 - o Assigned universities
6. Support approval-based permissions for sensitive actions.
7. Maintain a role-change audit trail.
8. Allow temporary access with an expiry date.
9. Prevent users from granting permissions higher than their own.
10. Support segregation of duties.

3.4 Dashboard and Analytics

The dashboard shall provide:

- Total leads
- Fresh leads
- Active applications
- Pending documents
- Submitted applications
- Offers received

- Unconditional offers
- Visa applications
- Enrolments
- Withdrawn or rejected applications
- Conversion percentages
- Weekly and monthly applications
- Team size and workload
- University distribution
- Country distribution
- Lead-source distribution
- Office performance
- Agent performance
- Counsellor performance
- Application ageing
- Pending reminders
- Upcoming deadlines
- Financial overview
- Commission forecast

Users shall be able to:

1. Filter dashboards by date, office, team, user, source, country, university, programme, intake, and application status.
2. Drill down from dashboard metrics to underlying records.
3. Save personal dashboard views.
4. Rearrange dashboard widgets.
5. Export dashboard data.
6. Compare current performance against previous periods.
7. View trends, targets, rankings, and exceptions.
8. Configure performance-score formulas.
9. View only data allowed by their permissions.

10. Receive alerts for unusual drops, spikes, delays, or missed targets.

3.5 Lead Management

The system shall:

1. Capture leads manually or automatically.
2. Accept leads from:
 - o Website forms
 - o Facebook and Instagram
 - o Google Ads
 - o Google Sheets
 - o CSV or Excel
 - o API
 - o WhatsApp
 - o Email
 - o Referral links
 - o Walk-ins
 - o Events and education fairs
3. Store:
 - o Personal details
 - o Contact details
 - o Nationality
 - o Country of residence
 - o Education history
 - o Programme interest
 - o Destination preference
 - o Budget
 - o Intake preference
 - o Lead source and campaign
4. Detect duplicate leads using email, phone, passport, or configurable matching rules.

5. Merge duplicate records.
6. Assign leads automatically or manually.
7. Support round-robin, workload-based, geographic, office-based, and programme-based assignment.
8. Maintain lead stages such as:
 - o New
 - o Contacted
 - o Qualified
 - o Counselling
 - o Documents pending
 - o Application initiated
 - o Converted
 - o Lost
 - o Unresponsive
9. Record lead score and conversion probability.
10. Schedule calls, meetings, follow-ups, and reminders.
11. Maintain complete lead activity history.
12. Record lost-lead reasons.
13. Support lead reactivation.
14. Allow bulk assignment, tagging, messaging, archiving, and status updates.
15. Track response time, first contact time, and lead ageing.
16. Support custom fields and tags.

3.6 Student and Applicant Profile

Each applicant profile shall include:

Personal Information

- Full legal name
- Preferred name
- Date of birth
- Gender

- Nationality
- Country of residence
- Passport and national ID details
- Marital status
- Disability or accessibility requirements
- Emergency contact

Contact Information

- Email
- Mobile number
- WhatsApp number
- Current and permanent address
- Preferred contact method

Academic Information

- School, college, and university history
- Qualification level
- Degree title
- Institution
- Country
- Attendance dates
- Completion date
- Grades, GPA, percentage, and classification
- Subjects studied
- Backlogs, gaps, and failed courses

English-Language Information

- IELTS
- PTE
- TOEFL
- Duolingo

- Medium-of-instruction evidence
- Test date, scores, and expiry

Employment Information

- Employer
- Job title
- Employment period
- Responsibilities
- Salary
- Gap explanations

Financial Information

- Sponsor
- Relationship to sponsor
- Available funds
- Bank details
- Scholarships
- Loan information

Additional Information

- Study gaps
- Previous visa refusals
- Immigration history
- Criminal or disciplinary declarations
- Medical declarations where legally justified
- Dependants
- Previous applications
- Personal statement
- Research proposal
- References

The system shall calculate profile completeness and identify missing or inconsistent data.

3.7 Application Management

The application module shall:

1. Allow one student to have multiple applications.
2. Generate unique application numbers.
3. Link each application to:
 - o Student
 - o University
 - o Campus
 - o Programme
 - o Intake
 - o Study mode
 - o Application type
 - o Office
 - o Team
 - o Counsellor
 - o Agent
4. Support configurable application workflows.
5. Maintain application stages, such as:
 - o Draft
 - o Initial review
 - o Documents pending
 - o Ready for submission
 - o Submitted
 - o University reviewing
 - o Additional information requested
 - o Conditional offer
 - o Unconditional offer
 - o Deposit pending

- o CAS or COE pending
 - o Visa preparation
 - o Visa submitted
 - o Visa granted
 - o Enrolled
 - o Deferred
 - o Withdrawn
 - o Rejected
6. Maintain status history with timestamps and responsible users.
 7. Allow users to configure custom statuses.
 8. Apply required-document checklists based on programme and stage.
 9. Highlight overdue applications.
 10. Allow applications to be reassigned.
 11. Support comments, internal notes, tags, and mentions.
 12. Maintain an application timeline.
 13. Support application copying for another programme or intake.
 14. Detect duplicate applications.
 15. Allow bulk status updates, reassignment, export, reminders, and communication.
 16. Support tabs and saved application views.
 17. Support user-defined columns.
 18. Allow applications to be archived and restored.
 19. Record rejection, withdrawal, cancellation, and deferral reasons.
 20. Support university-specific application forms.
 21. Allow internal approval before external submission.
 22. Lock approved fields after submission unless reopened by an authorized user.
 23. Maintain submission evidence, reference numbers, and acknowledgements.
 24. Support conditional-offer tracking and condition completion.
 25. Track deposits, tuition deadlines, CAS, visa, and enrolment milestones.

26. Support application checklists and progress percentages.

3.8 Document Management

The system shall:

1. Support upload through browser, mobile, email, API, and student portal.
2. Accept common file formats including PDF, image, document, and spreadsheet formats.
3. Categorize documents, including:
 - o Passport
 - o National ID
 - o Photograph
 - o Academic transcript
 - o Degree certificate
 - o English test
 - o CV
 - o Personal statement
 - o Reference letter
 - o Financial evidence
 - o Sponsor documents
 - o Visa document
 - o Offer letter
 - o CAS or COE
4. Apply programme-specific and country-specific document requirements.
5. Display received, missing, rejected, expired, and approved documents.
6. Support document verification.
7. Record verification comments and verified-by details.
8. Support document versioning.
9. Preserve the original uploaded file.
10. Support preview, download, rename, replace, archive, and delete permissions.
11. Extract text using OCR where appropriate.

12. Extract structured information using AI.
13. Detect:
 - Missing pages
 - Blurred files
 - Expired passports
 - Inconsistent names
 - Conflicting dates
 - Incomplete transcripts
14. Detect duplicate files.
15. Support electronic signature.
16. Support document expiry alerts.
17. Watermark downloaded documents where required.
18. Encrypt documents at rest and in transit.
19. Scan uploaded files for malware.
20. Maintain document access and download logs.
21. Allow secure document sharing through expiring links.
22. Support configurable retention and deletion policies.
23. Restrict highly sensitive documents to authorized roles.

3.9 University and Programme Management

The academic hierarchy shall support:

- Country
- City
- University
- Campus
- Faculty or school
- Department
- Subject area
- Degree type

- Programme
- Intake
- Study mode
- Session
- Currency
- Fee category
- Scholarship
- Document requirement
- Application deadline

The system shall:

1. Allow manual and bulk creation of academic records.
2. Import academic data through Excel or CSV.
3. Validate imports before committing records.
4. Show created, matched, skipped, duplicate, and failed records.
5. Provide downloadable error reports.
6. Make imports idempotent.
7. Support rollback of failed or incorrect imports.
8. Maintain import history.
9. Store programme information including:
 - o Title
 - o Award
 - o Duration
 - o Level
 - o Study mode
 - o Campus
 - o Tuition fee
 - o Application fee
 - o Deposit

- o Scholarship
 - o Entry requirements
 - o English-language requirements
 - o Work-experience requirements
 - o Portfolio requirements
 - o Start dates
 - o Deadlines
 - o Programme URL
10. Store country-specific entry requirements.
 11. Support programme versioning.
 12. Archive expired programmes and intakes.
 13. Support course-data feeds or APIs.
 14. Record the source and last verification date of programme data.
 15. Flag programme data requiring review.
 16. Support equivalency rules for academic qualifications.
 17. Support progression and pathway programmes.
 18. Support partner-specific commission arrangements.
 19. Support programme availability and capacity indicators.

3.10 Programme Search and Recommendation

A comprehensive system should include a programme-matching engine.

The system shall:

1. Allow users to search programmes by:
 - o Country
 - o University
 - o Subject
 - o Qualification
 - o Intake
 - o Fee

- o Scholarship
 - o Duration
 - o Study mode
 - o English requirement
 - o Academic requirement
- 2. Compare multiple programmes.
- 3. Allow programmes to be shortlisted.
- 4. Match applicants against entry requirements.
- 5. Show eligibility as:
 - o Likely eligible
 - o Potentially eligible
 - o Additional review required
 - o Unlikely eligible
- 6. Explain why a programme was recommended.
- 7. Identify missing eligibility information.
- 8. Allow counsellors to override recommendations with justification.
- 9. Avoid making final admissions decisions automatically.
- 10. Record recommendation history.
- 11. Support alternative and pathway recommendations.

3.11 AI Counsellor

The AI Counsellor shall:

1. Guide users through application creation.
2. Analyse uploaded academic, identity, and financial documents.
3. Extract applicant information.
4. Prepopulate application forms.
5. Identify missing documents.
6. Detect information conflicts.

7. Recommend suitable universities and programmes.
8. Draft:
 - o Personal statements
 - o Student summaries
 - o Application cover notes
 - o Follow-up messages
 - o Document requests
9. Summarize student profiles.
10. Generate application-readiness scores.
11. Explain AI recommendations.
12. Require human approval before submitting or changing critical data.
13. Clearly label AI-generated content.
14. Allow users to edit AI-generated content.
15. Maintain AI request and decision logs.
16. Prevent sensitive data from being used for unrelated model training without authorization.
17. Allow organizations to disable specific AI functions.
18. Apply usage limits and AI-credit tracking.
19. Detect low-confidence extraction and request human review.
20. Avoid presenting legal, immigration, or admission outcomes as guaranteed.

3.12 Communication Management

The system shall support:

- Email
- SMS
- WhatsApp
- Internal messaging
- Portal notifications
- Push notifications
- Automated reminders

Functional requirements include:

1. Send individual and bulk messages.
2. Use reusable message templates.
3. Support merge fields.
4. Maintain communication history.
5. Associate messages with a student, lead, or application.
6. Track delivered, opened, failed, and replied statuses where supported.
7. Support two-way communication.
8. Allow messages to be scheduled.
9. Trigger automated messages based on workflow events.
10. Support communication preferences and opt-outs.
11. Prevent duplicate or excessive automated messaging.
12. Allow counsellors to create follow-up tasks from messages.
13. Support attachments.
14. Support shared team inboxes.
15. Allow internal notes that are not visible to students or agents.
16. Support escalation of unanswered messages.
17. Store consent evidence for marketing communication.

3.13 Tasks, Calendar, Reminders, and Workflow Automation

The system shall:

1. Allow users to create tasks against leads, students, and applications.
2. Set owner, due date, priority, status, and reminder.
3. Support recurring tasks.
4. Display daily, weekly, and monthly calendars.
5. Synchronize with Google Calendar and Microsoft Outlook.
6. Allow appointment booking.
7. Support counsellor availability and meeting slots.
8. Trigger tasks automatically when:

- o A lead is created
 - o A document is uploaded
 - o An application changes stage
 - o A university requests information
 - o A deadline approaches
 - o A payment becomes due
9. Escalate overdue tasks.
 10. Notify team leaders of unresolved high-priority tasks.
 11. Support workflow rules based on conditions and actions.
 12. Allow administrators to configure workflow rules without coding.
 13. Prevent circular or conflicting workflow automation.
 14. Maintain workflow-execution logs.

3.14 Agent Management

The system shall:

1. Allow admins to add agents manually.
2. Support agent self-registration.
3. Review and approve pending agents.
4. Store:
 - o Legal business name
 - o Contact person
 - o Address
 - o Countries of operation
 - o Registration details
 - o Tax details
 - o Bank details
 - o Agreements
 - o Compliance documents
5. Assign institutions, territories, and programmes to agents.

6. Generate unique referral links or codes.
7. Allow agents to submit applications.
8. Restrict agents to their own students and applications.
9. Track agent leads, applications, conversion, enrolments, and commissions.
10. Support agent contracts and expiry alerts.
11. Support commission rules by university, programme, intake, or country.
12. Generate commission statements.
13. Allow commission disputes and approvals.
14. Suspend or deactivate agents.
15. Maintain agent performance ratings.
16. Support sub-agents where permitted.
17. Detect potential duplicate or fraudulent agent submissions.

3.15 Student Portal

The student portal shall allow applicants to:

1. Register and verify their account.
2. Complete their personal and academic profile.
3. Upload and replace documents.
4. View missing-document requirements.
5. Search and shortlist programmes.
6. Approve programme selections.
7. Review application information before submission.
8. Provide declarations and electronic signatures.
9. Track application status.
10. View university requests.
11. Download offer letters and other documents.
12. Pay application fees, deposits, or service fees.
13. View receipts and invoices.
14. Communicate with counsellors.

15. Schedule appointments.
16. Receive reminders.
17. Update contact information.
18. Submit feedback or complaints.
19. Request correction or deletion of personal data.
20. View consent and privacy settings.

3.16 University Partner Portal

Where universities are given portal access, they shall be able to:

1. View assigned applications.
2. Search and filter applications.
3. Review applicant profiles and documents.
4. Request additional documents.
5. Add internal or applicant-visible comments.
6. Update application status.
7. Upload offer letters and decisions.
8. Record scholarship decisions.
9. Export application data.
10. View source-agent and recruitment performance.
11. Manage programme and intake information subject to approval.
12. Access only authorized institutional data.

3.17 Form Creator

The form builder shall:

1. Allow drag-and-drop form creation.
2. Support:
 - o Text fields
 - o Number fields
 - o Email
 - o Phone

- o Date
 - o Dropdown
 - o Radio button
 - o Checkbox
 - o File upload
 - o Address
 - o Consent
 - o Signature
 - o Repeating education or employment sections
- 3. Support conditional fields.
- 4. Support validation rules.
- 5. Support multilingual forms.
- 6. Allow forms to be linked to a university, programme, campaign, or source.
- 7. Publish forms through:
 - o Public link
 - o Embedded code
 - o QR code
- 8. Apply organization branding.
- 9. Send submission confirmations.
- 10. Prevent spam through CAPTCHA and rate limiting.
- 11. Detect duplicates.
- 12. Automatically create leads or applications.
- 13. Track views, starts, completions, and drop-offs.
- 14. Allow form cloning and versioning.
- 15. Preserve the version used for each submission.
- 16. Allow forms to be unpublished without deleting prior responses.

3.18 Integrations

The system should integrate with:

- Facebook Lead Ads
- Instagram
- Google Sheets
- Website forms
- Gmail
- Microsoft Outlook
- WhatsApp Business
- SMS gateways
- Google Calendar
- Microsoft Calendar
- Cloud storage
- Payment gateways
- Accounting systems
- University APIs
- Identity-verification services
- Electronic-signature services
- Learning or student information systems
- Visa or immigration-status services where legally permitted

For each integration, the system shall:

1. Support secure OAuth or API credentials.
2. Show connection status.
3. Allow connection testing.
4. Maintain synchronization logs.
5. Show failed items.
6. Support retry and reprocessing.
7. Allow field mapping.
8. Prevent duplicate imports.
9. Notify admins when credentials expire.

10. Allow integrations to be disabled.
11. Log data sent to and received from third parties.
12. Support webhooks.
13. Validate webhook signatures.
14. Apply rate limiting and retry policies.

3.19 Reporting and Business Intelligence

Standard Reports

The system shall provide:

- Leads by source
- Lead conversion
- Applications by status
- Applications by university
- Applications by country
- Applications by intake
- Applications by office
- Applications by counsellor
- Applications by agent
- Offer rate
- Visa success rate
- Enrolment rate
- Rejection reasons
- Withdrawal reasons
- Pending-document ageing
- Application turnaround time
- Counsellor productivity
- Team workload
- Agent performance
- University performance

- Campaign performance
- Revenue and commissions
- Outstanding payments
- Forecasted enrolments
- Data-quality issues
- User activity
- Audit reports

Reporting Capabilities

The system shall:

1. Support pivot-style reports.
2. Support AI-assisted natural-language reporting.
3. Translate user questions into validated report queries.
4. Display filters and calculation logic used by AI reports.
5. Allow reports to be saved.
6. Allow report templates to be shared.
7. Schedule reports for email delivery.
8. Export reports to CSV, Excel, and PDF.
9. Support tables and visual charts.
10. Allow drill-down to source records.
11. Apply role-based data restrictions.
12. Support configurable KPIs.
13. Compare offices, teams, counsellors, and time periods.
14. Support report snapshots.
15. Prevent unauthorized exposure of personal data through reporting.
16. Limit AI reporting to available and authorized data.

3.20 Finance, Payments, and Commissions

The system shall:

1. Create invoices for:

- o Application fees
 - o Service charges
 - o Deposits
 - o Tuition payments
 - o Visa support
 - o Other configurable services
2. Accept online and offline payments.
 3. Record payment method, reference, date, currency, and amount.
 4. Generate receipts.
 5. Support discounts, scholarships, waivers, and refunds.
 6. Support multi-currency transactions.
 7. Store historical exchange rates used for calculations.
 8. Track unpaid and partially paid invoices.
 9. Send payment reminders.
 10. Reconcile gateway transactions.
 11. Calculate agent commissions.
 12. Support commission approvals.
 13. Generate agent statements.
 14. Track commissions as pending, eligible, approved, paid, disputed, or reversed.
 15. Record university receivables.
 16. Export financial records to accounting systems.
 17. Restrict financial information to authorized roles.
 18. Maintain complete financial audit logs.

3.21 Notifications

The system shall notify users about:

- New lead assignment
- New application assignment
- Missing documents

- Document rejection
- Upcoming deadline
- Overdue task
- New student message
- University request
- Offer received
- Payment due
- Visa milestone
- Agent approval
- Commission approval
- Integration failure
- Import completion
- Storage threshold
- AI-credit threshold
- Security events

Users shall be able to configure notification preferences by channel and event type.

3.22 Search

The system shall provide global search across:

- Students
- Leads
- Applications
- Universities
- Programmes
- Documents
- Agents
- Users
- Tasks
- Messages

- Reports

Search should support:

- Typo tolerance
- Partial matching
- Filters
- Recent searches
- Saved searches
- Permission-aware results
- Highlighted matches
- Advanced query conditions

Sensitive fields should be masked in search results where appropriate.

3.23 Import, Export, and Data Migration

The system shall:

1. Import data from Excel and CSV.
2. Provide downloadable templates.
3. Validate column mapping and formats.
4. Show a preview before import.
5. Support dry-run validation.
6. Detect duplicates.
7. Allow users to choose create, skip, update, or merge behaviour.
8. Produce detailed error reports.
9. Allow failed rows to be corrected and re-uploaded.
10. Maintain import history.
11. Allow authorized users to rollback imports.
12. Export permitted data.
13. Mask sensitive information in exports where required.
14. Add an export reason and approval workflow for bulk sensitive-data exports.
15. Log every export.

3.24 Data Quality Management

The system shall identify:

- Missing mandatory fields
- Invalid email or phone formats
- Duplicate students
- Duplicate applications
- Expired documents
- Inconsistent names
- Impossible or conflicting dates
- Invalid academic scores
- Missing programme information
- Stale university data
- Unassigned applications
- Applications without recent activity

A data-quality dashboard shall show issues, ownership, priority, and resolution status.

3.25 Audit Trail

The system shall log:

- Login and logout
- Failed login attempts
- Record creation
- Field changes
- Status changes
- Assignment changes
- Document upload, view, download, and deletion
- Exports and imports
- Role and permission changes
- AI-generated recommendations
- Communication

- Payment activity
- Agent commission actions
- Configuration changes
- Integration activity

Audit logs shall show who, what, when, where, previous value, new value, and reason where required.

Audit records shall not be editable by ordinary users.

3.26 Support and Case Management

The application should include:

1. Technical-support tickets.
2. Student complaints and service cases.
3. Ticket categories, priorities, ownership, and SLA.
4. Internal notes and attachments.
5. Escalation rules.
6. Status tracking.
7. Satisfaction rating.
8. Knowledge-base articles.
9. Frequently asked questions.
10. Support analytics.

3.27 Configuration and Master Data

Administrators shall configure:

- Countries
- Cities
- Nationalities
- Currencies
- Degree levels
- Qualification types
- Grading systems
- Subjects

- Intake periods
- Application types
- Application statuses
- Document types
- Lead sources
- Rejection reasons
- Withdrawal reasons
- Payment types
- Communication templates
- Workflow rules
- Custom fields
- Tags
- Holidays
- SLA thresholds
- Retention policies

Changes to critical master data should support approval and version history.

4. Non-Functional Requirements

4.1 Performance

1. Standard pages should load within **two seconds** under normal operating conditions.
2. Search results should appear within **two seconds** for commonly used queries.
3. Standard save and update operations should complete within **three seconds**.
4. Dashboard loading should complete within **five seconds**.
5. Large reports may be processed asynchronously.
6. Bulk imports should display progress.
7. Long-running operations should not block the user interface.
8. Pagination or virtualization should be used for large datasets.
9. The system should support caching for frequently accessed reference data.
10. Application performance should be monitored continuously.

4.2 Scalability

The architecture should support:

- Thousands of organizations
- Tens of thousands of concurrent users
- Millions of leads and applications
- Large document repositories
- Horizontal scaling
- Independent scaling of web, background-processing, reporting, search, storage, and AI services

The system should support queue-based processing for imports, notifications, OCR, AI analysis, and report generation.

4.3 Availability and Reliability

1. Target availability should be at least **99.9%**, excluding approved maintenance.
2. No single server failure should cause complete service interruption.
3. Critical background jobs should support automatic retry.
4. Failed integrations should enter a retry or exception queue.
5. Data writes should be transactional.
6. The system shall prevent partial application updates.
7. Health checks should monitor major system components.
8. Maintenance periods should be announced.
9. Users should see a meaningful service-status message during outages.

4.4 Security

The system shall:

1. Use HTTPS for all communication.
2. Encrypt sensitive data at rest.
3. Apply secure password hashing.
4. Support MFA.
5. Implement least-privilege access.
6. Protect against:

- o SQL injection
 - o Cross-site scripting
 - o Cross-site request forgery
 - o Broken access control
 - o Insecure file upload
 - o Session fixation
 - o Brute-force attacks
 - o API abuse
7. Use secure, HTTP-only, and same-site cookies.
 8. Apply rate limiting.
 9. Validate all inputs on both client and server.
 10. Use anti-malware scanning for files.
 11. Use secure secrets management.
 12. Rotate API keys and credentials.
 13. Mask sensitive data in logs.
 14. Conduct vulnerability scanning and penetration testing.
 15. Support immediate account and token revocation.
 16. Maintain security-event logs.
 17. Alert admins about suspicious login activity.
 18. Separate production, test, and development data.
 19. Prohibit use of live personal data in testing unless appropriately anonymized.
 20. Apply tenant isolation at both application and database-access levels.

4.5 Privacy and Data Protection

The system should support applicable privacy obligations, including:

1. Explicit consent collection.
2. Purpose-based data processing.
3. Data minimization.
4. Right of access.

5. Right of correction.
6. Right of deletion, subject to lawful retention.
7. Consent withdrawal.
8. Data-portability requests.
9. Marketing opt-out.
10. Configurable retention schedules.
11. Automated anonymization or deletion.
12. Data-processing agreements.
13. Third-party processor records.
14. Cross-border data-transfer controls.
15. Privacy notices by organization.
16. Data-breach management.
17. Data-subject request tracking.
18. Restricted access to highly sensitive data.
19. Privacy-aware test and analytics environments.

4.6 Backup and Disaster Recovery

1. Automated backups shall be taken regularly.
2. Backups shall be encrypted.
3. Backups shall be stored in geographically separate locations where appropriate.
4. Restore procedures shall be tested.
5. The system should define:
 - o Recovery Point Objective
 - o Recovery Time Objective
6. Documents and database records should be recoverable consistently.
7. Authorized admins should be able to view backup status.
8. Deletion of backups should require restricted privileges.
9. Business-continuity procedures should be documented.

A reasonable initial target would be an RPO of one hour and an RTO of four hours, subject to business needs.

4.7 Usability

1. Navigation should remain consistent throughout the application.
2. Frequently used operations should require minimal clicks.
3. Forms should provide inline validation.
4. Error messages should clearly explain what happened and how to correct it.
5. Unsaved changes should trigger a warning.
6. Long forms should support autosave.
7. Users should be able to resume incomplete work.
8. Bulk actions should show confirmation and impact.
9. Destructive actions should require confirmation.
10. The system should provide contextual help and onboarding guidance.
11. Empty states should guide the user toward the next action.
12. Terminology should be configurable for different markets.

4.8 Accessibility

The interface should target WCAG 2.2 AA compliance.

It should support:

- Keyboard navigation
- Screen readers
- Visible focus indicators
- Sufficient colour contrast
- Form labels
- Accessible error messages
- Alternative text for meaningful images
- Scalable text
- Reduced-motion settings
- Non-colour status indicators
- Accessible charts and tables

4.9 Responsiveness

The system shall work on:

- Desktop
- Laptop
- Tablet
- Mobile browser

Key student and agent workflows should be optimized for mobile devices, particularly document upload, status tracking, communication, and lead submission.

4.10 Compatibility

The application should support the latest stable versions of:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Apple Safari

The system should provide graceful degradation when an unsupported browser is used.

4.11 Maintainability

1. The application should use modular architecture.
2. Business logic should be separated from the user interface.
3. APIs should be versioned.
4. Code should follow agreed standards.
5. Automated unit, integration, security, and end-to-end tests should be maintained.
6. Database changes should use controlled migrations.
7. Configuration should be externalized.
8. System modules should be deployable with minimal disruption.
9. Technical documentation should be maintained.
10. Deprecated features should follow a formal removal process.

4.12 Observability

The system should provide:

- Centralized logs
- Performance metrics
- Error tracking
- Background-job monitoring
- Integration monitoring
- API monitoring
- Security-event monitoring
- Usage analytics
- Alerting
- Correlation identifiers for tracing a transaction across services

Logs must not expose passwords, tokens, financial details, or unnecessary personal data.

4.13 Localization

The system should support:

- Multiple languages
- Right-to-left languages
- Local date formats
- Local time zones
- Multiple currencies
- Country-specific address formats
- Country-specific academic grading systems
- Localized templates and notifications

All stored timestamps should use a consistent standard and display according to the user's configured time zone.

4.14 Data Integrity

1. Required relationships should be enforced.
2. Unique application and student identifiers should be maintained.
3. Referential integrity should be protected.

4. Duplicate-prevention rules should be configurable.
5. Historical records should not be overwritten without versioning.
6. Financial transactions should use precise decimal calculations.
7. Critical operations should be idempotent.
8. Concurrent updates should be safely handled.
9. Approved or submitted applications should have controlled editing.

4.15 API Requirements

The platform should expose secure APIs for authorized integration.

The APIs shall:

1. Use token-based authentication.
2. Apply scopes and permissions.
3. Be versioned.
4. Provide consistent errors.
5. Support pagination, filtering, and sorting.
6. Enforce rate limits.
7. Maintain request logs.
8. Support idempotency keys for create operations.
9. Offer webhooks for major events.
10. Provide developer documentation and a sandbox environment.
11. Mask or omit unauthorized fields.
12. Validate webhook signatures and retry failed deliveries.

5. Important Gaps Filled Beyond the Screenshots

The attached application presents a strong operational foundation, but the following functions are essential for a holistic solution:

1. Full lead-management lifecycle.
2. Student self-service portal.
3. University partner portal.

4. Document verification, versioning, expiry, and compliance controls.
5. Programme eligibility and recommendation engine.
6. Finance, payments, refunds, and agent commissions.
7. Configurable workflow automation.
8. Tasks, calendars, appointments, and SLA escalation.
9. Privacy, consent, retention, and data-subject request management.
10. Complete audit trail.
11. Data-quality dashboard.
12. Application checklists and readiness scoring.
13. Visa, deposit, CAS, and enrolment tracking.
14. Internal support and complaint management.
15. Commission and contract management for external agents.
16. Secure public APIs and webhooks.
17. AI governance and human approval.
18. Backup, disaster recovery, monitoring, and business continuity.
19. Accessibility and multilingual support.
20. Configurable role, field, office, and record-level permissions.

6. Recommended Application Modules

A comprehensive menu structure could be:

1. Dashboard
2. Leads
3. Students
4. Applications
5. Tasks and Calendar
6. Documents
7. Programme Search
8. AI Counsellor

9. Communications
10. Universities and Courses
11. Agents and Partners
12. Finance and Commissions
13. Forms and Campaigns
14. Reports and Analytics
15. Data Quality
16. Integrations
17. Organization and Offices
18. Teams, Roles, and Permissions
19. Support Cases
20. Audit Logs
21. Notifications
22. System Settings

7. Suggested Core Data Entities

The primary database entities should include:

- Organization
- Office
- User
- Role
- Permission
- Team
- Agent
- Agent Agreement
- Student
- Lead
- Lead Source

- Campaign
- Application
- Application Status
- Application History
- University
- Campus
- Faculty
- Programme
- Intake
- Entry Requirement
- Scholarship
- Document
- Document Type
- Document Verification
- Task
- Reminder
- Appointment
- Message
- Communication Template
- Form
- Form Submission
- Invoice
- Payment
- Refund
- Commission
- Workflow
- Workflow Execution
- Integration

- Import Job
- Export Job
- Notification
- Consent
- Support Ticket
- Audit Log
- AI Request
- AI Recommendation
- Subscription
- Usage Record

8. Recommended Delivery Priorities

Phase 1: Core CRM

- Authentication and roles
- Organization and office management
- Lead management
- Student profiles
- Application management
- University and programme master data
- Document upload
- Basic dashboards
- Tasks and reminders
- Import and export
- Audit logging

Phase 2: Operational Automation

- Student portal
- Agent portal
- Communication integrations

- Workflow automation
- Advanced document verification
- Programme eligibility matching
- Advanced reporting
- Calendar integrations
- Form builder

Phase 3: Advanced Platform

- AI Counsellor
- University portal
- Finance and commissions
- Visa and enrolment management
- Natural-language reporting
- Data-quality intelligence
- Mobile application
- Advanced APIs and partner ecosystem